

Automatic Streetlight Control and Fault Detection System

Project Report



University Logo

Names: N.Naagul
R.Prasanna
B.Sathursan

Group: Tech Nova

Module: EN1190

Index Numbers: 240445H
240525E
240602M

Date of Submission: June 30, 2026

Abstract

This project presents an automated streetlight control and fault detection system designed to minimize energy waste and speed up maintenance for municipal councils. Traditional streetlights often suffer from inefficient daytime operation and delayed fault reporting. To address this, the proposed system uses a central microcontroller integrated with natural light sensing for automated day/night switching, and current monitoring to detect bulb or wiring failures. When an anomaly is detected, a cellular module immediately alerts the responsible staff. The hardware architecture features built-in power regulation and automated load switching, packaged into a custom-designed Printed Circuit Board (PCB). Validated through stakeholder surveys, this low-cost, reliable system optimizes streetlight management, significantly reducing energy consumption while enhancing public safety.

List of Figures

1	System block diagram	4
2	Complete circuit schematic (Sheet1)	7
3	PCB layout (two-layer, top layer in red, bottom layer in blue)	8

List of Tables

1	System Specifications	11
2	LDR Switching Performance	11
3	Breakdown of Component Costs	13

Contents

Abstract	i
List of Figures	ii
List of Tables	iii
1 Introduction	1
1.1 Background	1
1.2 Problem Statement	1
1.3 Objectives	1
1.4 Scope	1
1.5 Methodology Overview	2
2 Background Research	2
3 Problem Validation	2
3.1 Questionnaire Design	2
3.2 Key Findings	2
3.3 Conclusion from Validation	3
4 Project Description	3
4.1 Detailed Methodology	3
4.1.1 Research and Problem Identification	3
4.1.2 Design Phase	3
4.1.3 Prototype Testing	3
4.1.4 PCB Design and Fabrication	4
4.1.5 System Testing and Documentation	4
4.2 Design and Development	4
4.2.1 System Architecture	4
4.2.2 Components	4
4.2.3 Circuit Design	6
4.2.4 PCB Design	7
4.3 Implementation	8
4.3.1 Abstract Model Design	8
4.3.2 Component Selection and Circuit Construction	8
4.3.3 Individual Component Testing	8
4.3.4 System Integration and Full Testing	9
4.3.5 Transition to PCB	9
4.3.6 Microcontroller Programming	9

4.4	Challenges and Solutions	10
4.4.1	Technical Challenges	10
4.4.2	Non-Technical Challenges	10
5	Results and Discussion	11
6	Marketing and Sales	12
6.1	Market Potential	12
6.1.1	Target Market	12
6.1.2	Market Need	12
6.2	Financial Analysis	12
6.2.1	Cost Breakdown	12
6.2.2	Pricing Strategy	13
6.2.3	Cost Justification	13
6.3	Return on Investment (ROI)	13
6.3.1	Economic Impact	13
6.3.2	Payback Period	14
6.4	Cost-Benefit Analysis	14
7	Conclusion	14
7.1	Summary of Findings	14
7.2	Implications	14
7.3	Future Work	14
8	Recommendations	15
8.1	Use a Stable Power Supply for the GSM Module	15
8.2	Add Averaging to Sensor Readings	15
8.3	Protect the Circuit from the Environment	15
8.4	Include Overvoltage and Overcurrent Protection	16
8.5	Test the GSM Coverage Before Installation	16
9	Individual Contributions	16
9.1	[Student 1 Name]	16
9.2	[Student 2 Name]	16
9.3	[Student 3 Name]	16
9.4	[Student 4 Name]	16
A	Appendices	18
A.1	Component Datasheets	18
A.2	Problem Validation Questionnaire	18
A.3	Acknowledgements	19

1 Introduction

1.1 Background

This project was carried out as part of the EN1190 module during our second semester. The main goal of the module was to design an electronic solution to a real problem that exists in society. After thinking about several options, we decided to focus on an issue that affects many people every day: the management of streetlights in urban and semi-urban areas.

Streetlights are essential for public safety at night. However, in many areas, they are not managed efficiently. Some lights remain on during the day, wasting electricity. Others fail and remain undetected for days, leaving roads in the dark and putting people at risk. These problems are mainly caused by the lack of an automated system that can control streetlights based on actual light conditions and detect faults as soon as they happen.

We decided to address this problem by designing a system that can automatically control streetlights and immediately report any faults to the relevant authority.

1.2 Problem Statement

Municipal councils in Sri Lanka spend a significant amount on electricity for streetlights. A large portion of this cost comes from lights that stay on during daytime, either because manual switching is unreliable or because time-based systems are not accurately configured. At the same time, when a streetlight fails, it is usually discovered only after a public complaint or a scheduled inspection. This delay means that faulty streetlights can remain broken for several days, creating safety risks and increasing maintenance costs. There is a clear need for a system that can control streetlights automatically and report faults without any human intervention.

1.3 Objectives

The main objectives of this project are:

- To design a circuit that automatically switches streetlights on and off based on the surrounding light level.
- To detect faults in streetlights by monitoring the current through the load.
- To send an SMS alert to the responsible person when a fault is detected.
- To build and test a working prototype on a custom PCB.

1.4 Scope

This project focuses on a single streetlight control unit that can be installed on an existing streetlight pole. The unit monitors the ambient light using an LDR sensor and controls the light using

a relay. It continuously monitors the current through the bulb and sends an SMS via a GSM module if the current reading indicates a fault. The system is powered directly from the mains supply using an on-board AC-to-DC converter.

1.5 Methodology Overview

The project began with a problem validation phase, where a structured questionnaire was distributed to people in municipal councils to confirm that the identified problems were genuine and that the proposed solution would be useful. Based on the validation results, the hardware design was started. The circuit was drawn in Altium and then transferred to a PCB layout. Individual components were tested separately before being combined into a full working system. The complete system was then tested to check whether it met the design objectives.

2 Background Research

3 Problem Validation

Before starting the design, we conducted a problem validation exercise. This was done using a structured questionnaire distributed in the municipal council area.

3.1 Questionnaire Design

The questionnaire contained 13 questions covering the following areas:

- The type of streetlights currently in use.
- How streetlights are currently switched on and off.
- How often lights remain on during the day.
- How often faults occur and what types of faults are common.
- How faults are currently detected and how long detection takes.
- The problems caused by delayed fault detection.
- Whether stakeholders believe an automated system would be useful.
- Their concerns about such a system.

3.2 Key Findings

The responses to the questionnaire confirmed the following:

- A significant number of respondents reported that streetlights staying on during the day-time was either a frequent or occasional occurrence in their area.

- Streetlight faults were reported to occur often, with bulb failure is being the most common fault types.
- Most faults were detected through public complaints and routine inspections, rather than through any automated system.
- In many cases, identifying a faulty streetlight took more than one day.
- The main concerns raised by respondents were installation cost, reliability, and maintenance complexity.

3.3 Conclusion from Validation

The validation survey confirmed that the problems we identified are real and significant. It also confirmed that the proposed solution would be welcomed by the people who manage and depend on streetlights. The concerns raised about cost and reliability guided our design decisions, particularly our choice to use low-cost, widely available components and a simple but robust circuit design.

4 Project Description

4.1 Detailed Methodology

4.1.1 Research and Problem Identification

The project began with background research to understand how streetlights are managed in Sri Lanka . We reviewed technical articles and product datasheets to understand what components were available and suitable for our application. We also consulted with people who manage streetlights to understand the practical challenges they face. This research gave us a clear picture of the problem and helped us define the requirements for our system.

4.1.2 Design Phase

Based on the research and validation findings, we began the circuit design. We first identified the key functions the system needed to perform: ambient light sensing, automatic switching, current monitoring, fault detection, and alert notification. We then selected appropriate components for each function and designed the circuit in a schematic editor.

4.1.3 Prototype Testing

After completing the schematic, we built a prototype to test each part of the circuit separately. This allowed us to check that each component was working correctly before combining everything into a single board.

4.1.4 PCB Design and Fabrication

Once the prototype was tested and any issues were resolved, we created a custom PCB layout based on the schematic. The PCB was designed to be compact and organised, with clear separation between high-voltage and low-voltage sections.

4.1.5 System Testing and Documentation

The complete system on the PCB was tested to verify that all functions worked together correctly. We then documented our process and results in this report.

4.2 Design and Development

4.2.1 System Architecture

The overall system consists of the following functional blocks:

1. **Power Supply:** Converts 230V AC mains to 5V DC and then to 3.3V DC.
2. **Microcontroller:** Reads sensor data, makes decisions, and controls outputs.
3. **Ambient Light Sensing:** Detects whether it is day or night.
4. **Relay Switching:** Turns the streetlight on or off.
5. **Current Monitoring:** Measures the current through the streetlight.
6. **GSM Communication:** Sends SMS alerts when a fault is detected.
7. **Programming Interface:** Allows the microcontroller firmware to be updated.

A block diagram showing these functional blocks and their interconnections is presented in Figure 1.

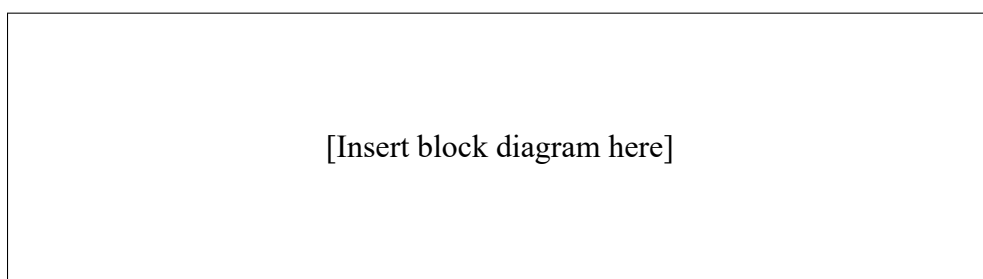


Figure 1: System block diagram

4.2.2 Components

ESP32-WROOM-32E Microcontroller The ESP32-WROOM-32E is used as the central processing unit of the system. It was chosen because it is powerful, widely available, affordable,

and has built-in support for both Wi-Fi and Bluetooth, which provides room for future expansion. It operates at 3.3V and has a large number of general-purpose input/output (GPIO) pins, which is needed for connecting all the peripheral components.

Key features relevant to this project include its analog input pins for reading the LDR and current sensor outputs, its digital output pins for controlling the relay, and its UART interface for communicating with the GSM module. Boot and reset buttons are included on the board, and a programming header (J3) is provided to allow firmware to be uploaded easily via a four-pin connector.

HLK-20M05 AC-to-DC Power Module The HLK-20M05 is a compact, enclosed power module that converts 230V AC mains voltage to a regulated 5V DC output with a rated current of up to 4A. This module was selected because it is simple to use, does not require any external components, and is enclosed for electrical safety. It directly powers the relay module and the AMS1117-3.3 voltage regulator.

The input connectors J_IN1 (AC live and neutral in) and J_OUT1 (load live and neutral out) are screw-terminal connectors of type KF301-2P, which are suitable for mains-level wiring.

AMS1117-3.3 Voltage Regulator The AMS1117-3.3 (U2) is a linear voltage regulator that takes the 5V DC output from the HLK-20M05 and produces a stable 3.3V DC output for the ESP32 and other 3.3V components. It is a simple three-terminal device with input, output, and ground/adjust pins. A bypass capacitor (C10, 10 μ F) is placed on the output to improve stability.

ACS712 Current Sensor The ACS712 (J21) is a Hall-effect based current sensor module. It measures the AC current flowing through the streetlight by producing an output voltage that is proportional to the current. This output is connected to one of the analog input pins of the ESP32 (IO_34). The microcontroller reads this voltage and calculates the current. If the measured current falls outside the expected range, this indicates a fault such as a bulb failure or an open circuit, and the system sends an alert.

Two pull-down resistors (R1, 10k Ω and R2, 22k Ω) form a voltage divider on the output of the ACS712 to scale the signal to a range suitable for the ESP32's ADC input. A bypass capacitor (C4, 10 nF) is also included to filter noise.

LDR (Light Dependent Resistor) The LDR (R6, labelled 02LDR1) is used to sense the ambient light level. It is connected in a voltage divider configuration with a fixed resistor (R7, 10k Ω). The midpoint voltage of this divider is connected to an analog input pin of the ESP32 (IO_35). When it is dark, the resistance of the LDR increases, causing the voltage at the midpoint to change. The microcontroller reads this voltage and determines whether the streetlight should be on or off.

Relay Module (K1 SRD-05VDC-SL-C) The relay (K1) is used to switch the mains supply to the streetlight. It is a 5V single-pole double-throw (SPDT) relay. The coil of the relay is driven by a 2N2222 NPN transistor (Q1), which in turn is controlled by a GPIO pin of the ESP32 (IO_26). A 1k Ω base resistor (R3) limits the current from the ESP32 pin into the transistor base. A 1N4007 flyback diode (D1) is connected in parallel with the relay coil to protect the transistor from the voltage spike that occurs when the relay is de-energised.

EVB GSM 800L Module The GSM module (connected via J2) is used to send SMS alerts to the maintenance team when a fault is detected. It communicates with the ESP32 over a UART serial interface using pins IO_16 (RX of ESP32) and IO_17 (TX of ESP32). The module operates on a 5V supply from the HLK-20M05, with 100 μ F (C6) and 1000 μ F (C7) bulk capacitors to handle the high current demand during transmission. IO_5 is used as an additional control signal to the GSM module.

Supporting Passive Components Several passive components are used throughout the circuit to ensure stable and reliable operation:

- **Decoupling capacitors** (C5, C9: 100 nF) are placed near the power supply pins of the ESP32 to filter out high-frequency noise.
- **Pull-up and pull-down resistors** (R4, R5: 10k Ω) are connected to the EN and IO_0 pins of the ESP32. These ensure the microcontroller starts up in the correct state and that the boot mode is set properly.
- **Boot and reset buttons** (S2, S3) allow manual control of the ESP32 boot and reset functions, which are needed during programming.

4.2.3 Circuit Design

The circuit was designed using a schematic editor. The schematic is divided into clearly labelled functional blocks:

- **230V to 5V Converter:** Contains the HLK-20M05 module and the input/output connectors.
- **5V to 3.3V Converter:** Contains the AMS1117-3.3 regulator and its supporting capacitors.
- **Relay Module:** Contains the relay K1, transistor Q1, diode D1, and resistor R3.
- **ACS712:** Contains the current sensor and its supporting resistors and capacitor.
- **LDR:** Contains the LDR and voltage divider resistor.
- **EVB GSM 800L:** Contains the GSM module connector and bulk capacitors.

- **ESP32-WROOM-32E:** The main microcontroller with all its connections.
- **Programming Header:** A four-pin connector for serial programming.

The complete schematic is shown in Figure 2.

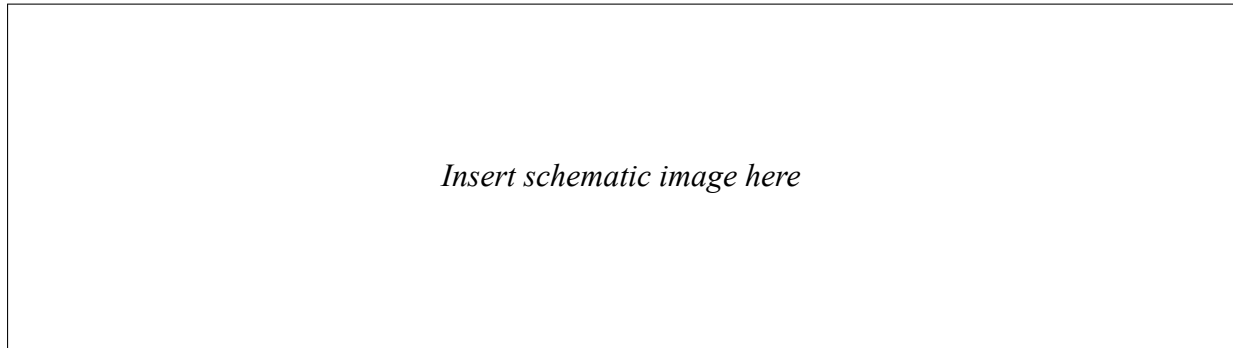


Figure 2: Complete circuit schematic (Sheet1)

4.2.4 PCB Design

The PCB was designed based on the schematic. Several important considerations were taken into account during the layout process.

Layer Strategy A two-layer PCB was used. The top layer (shown in red in Figure 3) carries most of the signal and power traces. The bottom layer (shown in blue) is used for return paths and some crossing routes. Vias are used where a trace needs to change layers.

Trace Width For low-current signal traces, a standard trace width was used to minimise board area. For power supply traces carrying higher currents (especially the 5V supply line and the relay output), wider traces were used to reduce resistive voltage drops and prevent overheating.

Component Placement Components were grouped according to their function and placed close to each other to keep trace lengths short. The high-voltage section (HLK-20M05, relay, and the mains connectors) was kept physically separated from the low-voltage section (ESP32, sensors, and GSM module) to reduce the risk of interference and to meet basic safety requirements.

Safety Clearance An adequate clearance was maintained between mains-voltage traces and low-voltage traces to prevent accidental short circuits and to comply with basic electrical safety standards.

The completed PCB layout is shown in Figure 3.

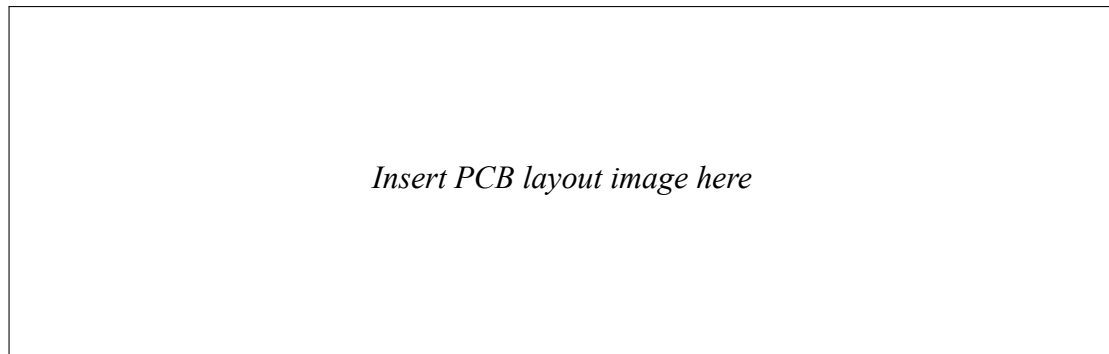


Figure 3: PCB layout (two-layer, top layer in red, bottom layer in blue)

4.3 Implementation

4.3.1 Abstract Model Design

The first step was to draw a simple block diagram of the system, showing the main functional blocks and how they connect to each other. This helped the team agree on the overall architecture before starting the detailed design.

4.3.2 Component Selection and Circuit Construction

After agreeing on the block diagram, the team researched and selected appropriate components for each block. The selection was based on availability, cost, operating voltage, and ease of use. Once the components were selected, the schematic was drawn. Each component was added to the schematic one block at a time, starting with the power supply, then the microcontroller, and then the peripheral components.

4.3.3 Individual Component Testing

Before combining everything on a single board, each component was tested individually using a breadboard prototype. This included:

- Testing the LDR voltage divider to verify that the output voltage changed correctly with varying light levels.
- Testing the ACS712 by passing a known current and reading the output voltage.
- Testing the relay and transistor driver by driving the relay coil from a microcontroller GPIO pin.
- Testing the GSM module by sending AT commands over serial and verifying that an SMS was received.
- Testing the ESP32 with basic firmware to confirm that all GPIO pins were working correctly.

4.3.4 System Integration and Full Testing

After individual component testing, all parts were connected together and the complete firmware was uploaded to the ESP32. The system was tested as a whole to verify that:

- The relay switched on when the LDR reading indicated darkness and switched off when light was detected.
- The ACS712 correctly detected both normal operating current and fault conditions (simulated by disconnecting the load).
- An SMS was sent when a fault was detected.
- The system recovered correctly after a fault was cleared.

4.3.5 Transition to PCB

Once the prototype was confirmed to be working correctly, the PCB was designed and fabricated. After receiving the PCB, components were soldered onto it and the full system was re-tested on the PCB to confirm that the PCB layout had no errors.

4.3.6 Microcontroller Programming

The ESP32 was programmed using the Arduino IDE with the ESP32 board support package. The firmware was structured as follows:

1. **Initialisation:** Configure all GPIO pins, start serial communication with the GSM module, and set initial output states.
2. **Main loop:** Continuously read the LDR sensor and the ACS712 sensor.
3. **Light control:** If the LDR reading indicates darkness, turn the relay on (streetlight on). If the LDR reading indicates daylight, turn the relay off.
4. **Fault detection:** If the relay is on but the ACS712 reading shows no current (or current below a threshold), a fault is assumed. An SMS is sent to the configured phone number with a fault alert message.
5. **Alert management:** To avoid sending repeated SMS messages for the same fault, the system tracks whether a fault alert has already been sent and only sends a new alert when the fault condition changes.

4.4 Challenges and Solutions

4.4.1 Technical Challenges

GSM Module Power Supply The EVB GSM 800L module requires a stable supply capable of delivering high peak currents during SMS transmission. In early testing, the module was resetting during transmission due to insufficient current supply. This was resolved by adding large bulk capacitors (100 μ F and 1000 μ F) close to the module's power supply pins, which store charge and supply the peak current needed during bursts of transmission activity.

ACS712 Noise The output of the ACS712 contains some noise, which caused occasional false fault readings. This was addressed by averaging several ADC readings in the firmware before making a fault decision, which effectively filtered out the noise.

LDR Threshold Sensitivity The threshold light level at which the relay should switch was difficult to set correctly, because the ambient light level varies with weather and location. We resolved this by making the threshold value a configurable parameter in the firmware, which can be adjusted without changing the hardware.

Mains Voltage Safety Working with 230V mains voltage required careful attention to safety during testing. We took precautions by testing the high-voltage section only after confirming that the low-voltage section was working correctly, and by using appropriate insulation and safety practices throughout.

4.4.2 Non-Technical Challenges

Component Availability Some components were not available locally and had to be ordered online, which caused delays. We addressed this by ordering components early and keeping spare parts for critical components.

Budget Management The cost of components exceeded our initial estimate, primarily due to the cost of the GSM module and the PCB fabrication. We managed this by reducing the number of prototype iterations and carefully planning the PCB layout to avoid the need for redesigns.

Team Coordination Coordinating the work among team members was at times challenging, particularly when some members were working on different aspects of the project simultaneously. We addressed this by holding regular meetings and using a shared document to track progress.

5 Results and Discussion

The complete system was assembled and tested. The performance of each functional block was evaluated, and the results are summarised in Table 1 and Table 2.

Table 1: System Specifications

Parameter	Details
Main controller	ESP32-WROOM-32E (16MB Flash)
Supply voltage (mains input)	230V AC
Regulated supply (to ESP32)	3.3V DC (via AMS1117-3.3)
Regulated supply (to relay, GSM)	5V DC (via HLK-20M05)
Light sensor	LDR (02LDR1) with 10k Ω pull-up
Current sensor	ACS712
Fault alert method	SMS via EVB GSM 800L
Relay type	SRD-05VDC-SL-C (SPDT)
Switching transistor	2N2222 NPN
Flyback protection	1N4007 diode
PCB layers	2

Table 2: LDR Switching Performance

Ambient Light Level	LDR Output (V)	Relay State
Bright daylight	>2.5	OFF
Overcast / dim	1.5 – 2.5	Configurable
Night / darkness	<1.5	ON

The system correctly switched the relay on when the LDR reading fell below the darkness threshold and switched it off when the reading rose above the daylight threshold. The ACS712 current sensor successfully distinguished between normal operation (relay on, bulb drawing current) and fault conditions (relay on, but no current flowing, indicating bulb failure or open circuit). When a fault was simulated by disconnecting the load while the relay was on, the ESP32 detected the fault and the GSM module sent an SMS alert within approximately 5 to 10 seconds.

The complete schematic, with all components and connections clearly labelled, is shown in Figure 2. The PCB layout, showing the physical arrangement of components on the board, is shown in Figure 3.

Overall, the system met all of the design objectives. It provides automatic day/night switching, real-time current monitoring for fault detection, and immediate SMS notification when a fault

occurs. The total component cost was kept low, making the system suitable for deployment by municipal councils with limited budgets.

6 Marketing and Sales

6.1 Market Potential

6.1.1 Target Market

The primary target market for this system is local government bodies in Sri Lanka, specifically municipal councils, urban councils, and pradeshiya sabhas that are responsible for the maintenance and operation of streetlights. There are hundreds of such local government bodies across Sri Lanka, and together they manage tens of thousands of streetlights. This represents a very large potential market.

6.1.2 Market Need

The problem validation survey confirmed that inefficient streetlight management and delayed fault detection are widespread problems. All of the organisations that manage streetlights would potentially benefit from this system.

6.2 Financial Analysis

6.2.1 Cost Breakdown

Table 3 shows the approximate cost of the components used to build one unit of the system.

Table 3: Breakdown of Component Costs

Component	Approximate Price (LKR)
ESP32-WROOM-32E module	1,200
HLK-20M05 power module	850
AMS1117-3.3 voltage regulator	50
ACS712 current sensor module	350
LDR sensor	30
SRD-05VDC-SL-C relay	120
2N2222 transistor	20
1N4007 diode	10
EVB GSM 800L module	1,500
Resistors and capacitors	150
KF301-2P screw terminals	80
PCB fabrication	1,800
Miscellaneous (wires, SIM, etc.)	500
Total	6,660

6.2.2 Pricing Strategy

Based on the component cost, the system can be priced between 8,000 and 10,000 LKR per unit. This price point covers the material cost, assembly labour, and a small profit margin. Bulk orders from councils could bring the per-unit price down further.

6.2.3 Cost Justification

The energy savings alone from eliminating daytime operation can justify the cost of the system. A single streetlight left on unnecessarily for 8 hours per day consumes a significant amount of energy over a month. Multiplied across hundreds of streetlights, the total saving can pay back the investment in the system within a few months.

6.3 Return on Investment (ROI)

6.3.1 Economic Impact

Municipal councils currently spend money on electricity wasted by lights that run during the day, and on maintenance visits to discover and repair faults that could be detected automatically. By reducing energy waste and enabling faster fault response, the system can produce significant cost savings for the council.

6.3.2 Payback Period

Given the energy and maintenance savings, the payback period for the system is estimated to be less than one year for most councils. After that, the system continues to save money for as long as it remains in operation.

6.4 Cost-Benefit Analysis

The system provides a favourable cost-benefit ratio. The initial investment is low compared to the long-term savings from reduced energy consumption and maintenance costs. The low-cost components used in the design also mean that repairs and replacements are inexpensive. The durability of the PCB-based design ensures that the system remains operational for many years with minimal maintenance.

7 Conclusion

7.1 Summary of Findings

This project successfully designed and built a prototype of an automatic streetlight control and fault detection system. The system uses an ESP32-WROOM-32E microcontroller connected to an LDR sensor for ambient light detection, an ACS712 current sensor for fault detection, a relay module for switching the streetlight, and a GSM module for sending SMS alerts. All components were powered from the mains supply through an HLK-20M05 power module and an AMS1117-3.3 voltage regulator. The complete circuit was drawn in a schematic editor and implemented on a custom two-layer PCB.

Testing confirmed that the system correctly switches the streetlight on at night and off during the day, and that it can detect bulb faults and send an SMS alert within a few seconds. The total component cost was kept below 7,000 LKR, making the system affordable for small local government bodies.

7.2 Implications

This project shows that it is possible to build an effective, affordable, and automated streetlight management system using low-cost, widely available components. If deployed at scale, such a system could significantly reduce the energy wasted on daytime streetlight operation and reduce the delay between fault occurrence and fault repair. Both of these outcomes would have a positive impact on public safety and on the budgets of local government bodies.

7.3 Future Work

The following improvements are planned for future versions of the system:

- **Wi-Fi / Cloud Integration:** The ESP32 already has built-in Wi-Fi capability. A future version of the firmware could send fault alerts and operational data to a cloud-based dashboard, allowing a single operator to monitor all streetlights in an area from one screen.
- **Dimming Control:** Instead of simply switching the light on or off, the system could be extended to dim the light during low-traffic hours using pulse-width modulation (PWM), further reducing energy consumption.
- **Solar Power:** For areas where mains power is unreliable, the system could be adapted to run from a solar panel and battery.
- **Multiple Fault Types:** The current system detects open-circuit faults (bulb failure). Future versions could also detect short-circuit faults and over-current conditions by adding more sophisticated signal processing to the ACS712 readings.
- **GPS Location Reporting:** Adding a GPS module would allow the SMS alert to include the location of the faulty streetlight, making it easier for maintenance teams to find and fix the problem quickly.

8 Recommendations

Based on the experience gained during this project, we make the following recommendations for anyone looking to build or extend a similar system.

8.1 Use a Stable Power Supply for the GSM Module

The GSM module is the most power-hungry component in the system. A power supply that cannot deliver enough current will cause the module to reset or fail during SMS transmission. We recommend using large bulk capacitors (at least 1000 μF) close to the module's power pins and ensuring that the supply can deliver at least 2A peak current.

8.2 Add Averaging to Sensor Readings

Both the LDR and the ACS712 produce noisy readings. Using a simple moving average (taking several readings and computing their mean) significantly reduces false triggers and improves the reliability of the system.

8.3 Protect the Circuit from the Environment

A streetlight controller will be installed outdoors and must withstand heat, humidity, and dust. We recommend placing the PCB in a sealed enclosure with an IP54 or better rating. Cable glands should be used for all wiring entries to prevent water ingress.

8.4 Include Overvoltage and Overcurrent Protection

Mains-connected circuits can be exposed to voltage surges and transients. We recommend adding a metal oxide varistor (MOV) across the mains input to protect against voltage surges, and a fuse on the mains input to protect against overcurrent faults.

8.5 Test the GSM Coverage Before Installation

The GSM module requires adequate mobile network coverage to send SMS alerts. Before installing the system, the GSM signal strength at the intended installation location should be checked to ensure that alerts will be delivered reliably.

9 Individual Contributions

9.1 [Student 1 Name]

[Student 1] was responsible for the overall circuit schematic design. This included selecting components, drawing the schematic in the design tool, and verifying that all connections were correct. [Student 1] also designed and tested the relay switching circuit and the power supply section.

9.2 [Student 2 Name]

[Student 2] was responsible for the PCB layout. This included placing components on the board, routing traces, and ensuring that the layout met safety and manufacturing requirements. [Student 2] also managed the PCB fabrication order.

9.3 [Student 3 Name]

[Student 3] was responsible for the microcontroller firmware. This included writing the Arduino code for the LDR-based switching, the ACS712 fault detection, and the GSM SMS sending. [Student 3] also carried out the component-level testing of each hardware block.

9.4 [Student 4 Name]

[Student 4] was responsible for the problem validation survey. This included designing the questionnaire, distributing it to respondents, analysing the results, and documenting the findings. [Student 4] also led the preparation of this project report.

References

- [1] Espressif Systems. *ESP32-WROOM-32E Datasheet*, 2022. Available at: https://www.espressif.com/sites/default/files/documentation/esp32-wroom-32e_esp32-wroom-32ue_datasheet_en.pdf
- [2] Allegro MicroSystems. *ACS712 Fully Integrated, Hall Effect-Based Linear Current Sensor Datasheet*, 2020. Available at: <https://www.allegromicro.com/en/products/sense/current-sensor-ics/zero-to-fifty-amp-integrated-conductor-sensor-ics/acs712>
- [3] Hi-Link Electronics. *HLK-20M05 Power Module Datasheet*. Available at: <https://www.hlktech.net>
- [4] Advanced Monolithic Systems. *AMS1117 1A Low Dropout Voltage Regulator Datasheet*. Available at: <http://www.advanced-monolithic.com/pdf/ds1117.pdf>
- [5] SIMCOM. *SIM800L Hardware Design Manual*, 2015. Available at: https://www.makerhero.com/img/files/download/Datasheet_SIM800L.pdf
- [6] Songle. *SRD-05VDC-SL-C Relay Datasheet*. Available at: <https://components101.com/switches/5v-single-channel-relay-module-pinout-features-applications-working->
- [7] ON Semiconductor. *2N2222 NPN General-Purpose Amplifier Datasheet*, 2014. Available at: <https://www.onsemi.com/pdf/datasheet/p2n2222a-d.pdf>
- [8] Vishay. *1N4007 General Purpose Rectifier Datasheet*, 2021. Available at: <https://www.vishay.com/docs/88503/1n4007.pdf>

A Appendices

A.1 Component Datasheets

- ESP32-WROOM-32E Datasheet: https://www.espressif.com/sites/default/files/documentation/esp32-wroom-32e_esp32-wroom-32ue_datasheet_en.pdf
- ACS712 Datasheet: <https://www.allegromicro.com/en/products/sense/current-sensor-ics/acs712>
- HLK-20M05 Datasheet: <https://www.hlktech.net>
- AMS1117-3.3 Datasheet: <http://www.advanced-monolithic.com/pdf/ds1117.pdf>
- SIM800L Datasheet: https://www.makerhero.com/img/files/download/Datasheet_SIM800L.pdf
- SRD-05VDC-SL-C Relay: <https://components101.com/switches/5v-single-channel-relay->
- 2N2222 Transistor: <https://www.onsemi.com/pdf/datasheet/p2n2222a-d.pdf>
- 1N4007 Diode: <https://www.vishay.com/docs/88503/1n4007.pdf>

A.2 Problem Validation Questionnaire

The following questionnaire was used to validate the problem before starting the design phase.

Automatic Streetlight Control and Fault Detection System — Problem Validation Questionnaire

EN1190 Engineering Design Project, University of Moratuwa

1. What type of streetlights are mainly used in your area?
 - a. LED
 - b. Sodium Vapour
 - c. CFL
 - d. Mixed
2. How are streetlights currently switched ON and OFF?
 - a. Manual Switching
 - b. Time-based system
 - c. LDR sensor
 - d. Centralised Control
3. How often do you observe streetlights remaining ON during daytime?
 - a. Frequently
 - b. Occasionally
 - c. Rarely
 - d. Never
4. How often do streetlight faults occur in your area?
 - a. Very often
 - b. Often
 - c. Rarely
 - d. Very Rarely
5. What are the common types of faults experienced?

- a. Bulb failure b. Power supply failure c. Wiring damage d. Control unit failure
e. Other
- 6. How are streetlight faults usually detected?
 - a. Public complaints b. Routine inspection c. Maintenance staff observation d. No fixed method
- 7. On average, how long does it take to identify a faulty streetlight?
 - a. Within 24 hours b. 1–3 days c. More than 3 days
- 8. What problems arise due to delayed fault detection?
 - a. Increased energy wastage b. Public safety concerns c. Increased maintenance cost d. Public complaints
- 9. Is energy wastage due to streetlights a concern for the council?
 - a. Yes b. No c. Not Sure
- 10. Would an automatic ON/OFF control system be beneficial?
 - a. Yes b. No c. Maybe
- 11. Would a system that automatically detects and reports streetlight faults be useful?
 - a. Very Useful b. Useful c. Not Useful
- 12. What are the main concerns regarding such a system?
 - a. Installation cost b. Maintenance complexity c. Reliability d. Safety e. Other
- 13. Do you have any suggestions or comments regarding improving streetlight management?

A.3 Acknowledgements

We sincerely thank the lab staff at the Analog Lab for their support and for accommodating our lab sessions. We also thank the administrative staff at ENTC for providing the equipment and resources needed for our experiments. We are grateful to Dr. Pasqual and our fellow batchmates for their guidance and encouragement throughout the project.